Sample question:

Suppose that Alice's RSA public key is (N, e) = (33,3) and her private key is d = 7.

a. If Bob encrypts the message M = 19 using Alice's public key, what is the ciphertext CI Show that Alice can decrypt C to obtain M.

Answer:

🔐 RSA Encryption & Decryption

Encryption: $C = M^e \bmod N$

Decryption: $M = C^d \bmod N$

📨 Encryption:

$C = M^e \bmod N$
$= 19^3 \bmod 33$
$= 6859 \bmod 33$
$= 28$

👉 **C = 28**

🔓 Decryption:

$M = C^d \bmod N$
$= 28^7 \bmod 33$
$= 28^3 * 28^3 * 28 \bmod 33$
$= 7 * 7 * 28$
$= 19$

✅ Final Answer:

M = 19

Sample question:

Suppose that for the knapsack cryptosystem, the super increasing knapsack is (3,5,12,23) with n = 47 and m = 6.

a. Give the public and private keys.
b. Encrypt the message M = 1110 (given in binary). Give your result in decimal.

Answer:

□ **Given:**

☞ Superincreasing knapsack (SIK):
(3, 5, 12, 23)

☞ n = 47, m = 6

☞ Message: **M = 1110 (binary)**

🅰 **(a) Public Key & Private Key**

🔐 Step 1: Private Key

☞ Rule:
Private key =
✔ SIK + m + n

☞ তাই:

✅ **Private Key = (3, 5, 12, 23), m = 6, n = 47**

🔓 Step 2: Public Key বের করি

☞ Formula:
W' = W × m mod n

এক এক করে করি:

☞ 3 × 6 = 18 mod 47 = **18**
☞ 5 × 6 = 30 mod 47 = **30**
☞ 12 × 6 = 72 mod 47 = **25**
☞ 23 × 6 = 138 mod 47 = **44**

✅ **Public Key = (18, 30, 25, 44)**

🅱 **(b) Encryption**

🔐 Step 3: Binary message বুঝো

☞ M = 1110

মানে:

Bit	Weight
1	18
1	30
1	25
0	44

☞ শুধু যেগুলা 1 আছে সেগুলা যোগ করো:

= 18 + 30 + 25
= **73**

✅ **Final Answer:**

✔ (a)

- Private Key: **(3, 5, 12, 23), m = 6, n = 47**
- Public Key: **(18, 30, 25, 44)**

✔ (b)
☞ **Ciphertext =** 73 (decimal)

◆ What is Cryptography?

Cryptography is the practice of transforming information (messages) so that **only authorized people can understand it**.

◆ Main Functions:

- **Encryption:** Convert **plaintext** → **ciphertext**
- **Decryption:** Convert **ciphertext** → **plaintext**

◆ **Purpose of Cryptography:**

1. **Confidentiality:** Keep the information secret
2. **Integrity:** Ensure data is not altered
3. **Authentication:** Verify the sender’s identity
4. **Non-repudiation:** Prevent the sender from denying sending the message

🎒 **Knapsack Problem কী?**

Knapsack problem হলো একটি **subset sum problem**।

☞ Definition:
দেওয়া থাকে কিছু weight:
$W_0, W_1, \ldots, W_{n-1}$
এবং একটি sum S

What is RSA?

RSA is a **public-key cryptography algorithm**, which means it uses **two keys**:

1. **Public Key** – for encryption (anyone can use it)
2. **Private Key** – for decryption (only the owner can use it)

It’s called the **gold standard** of public-key crypto because it’s widely used for secure communication.